



# RADIUS vs TACACS+ — Assignment

## Question 1

Create a comparison table listing at least 4 key differences between RADIUS and TACACS+ features, focusing on protocol type, encryption, ports used, and command authorization.

### Answer

**RADIUS (Remote Authentication Dial-In User Service)** and **TACACS+ (Terminal Access Controller Access-Control System Plus)** are AAA protocols used to control access to network resources.

| Feature                  | ● RADIUS                                                                        | ● TACACS+                                                            |
|--------------------------|---------------------------------------------------------------------------------|----------------------------------------------------------------------|
| 1. Protocol Type         | Uses <b>UDP</b>                                                                 | Uses <b>TCP</b>                                                      |
| 2. Encryption            | Encrypts the user's password; other parts of the packet are not fully encrypted | Encrypts the <b>entire TACACS+ packet body</b>                       |
| 3. Ports Used            | UDP <b>1812</b> for authentication and <b>1813</b> for accounting               | TCP <b>49</b>                                                        |
| 4. Command Authorization | Limited support for detailed device-command authorization                       | Provides <b>separate and detailed command authorization</b>          |
| 5. AAA Separation        | Authentication and authorization are commonly combined in the protocol exchange | Authentication, authorization, and accounting are handled separately |
| 6. Common Use            | Commonly used for <b>Wi-Fi, VPN, and network access authentication</b>          | Commonly used for <b>network-device administration</b>               |

### ★ Key Difference

**RADIUS** is commonly used to authenticate network users, while **TACACS+** is particularly useful for controlling and auditing administrators who manage network devices.

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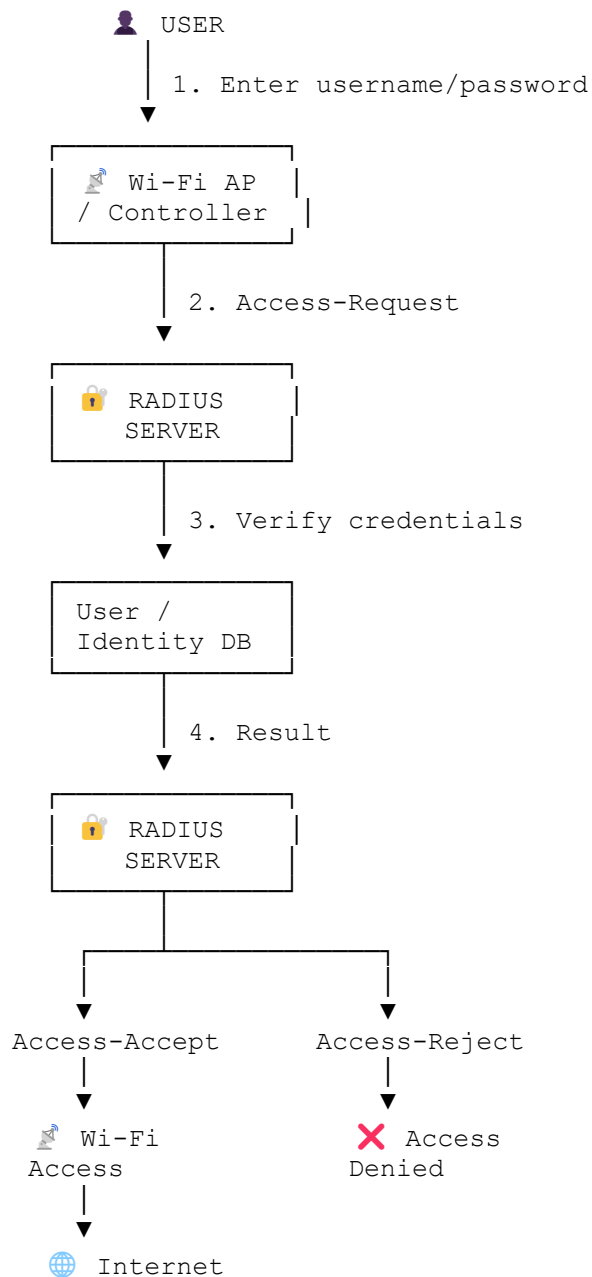
## Question 2

Draw a simple flowchart or sequence diagram that illustrates the **RADIUS** authentication process for a user trying to log in to a **Wi-Fi** network (like at a coffee shop or college campus).

## Answer



## RADIUS Wi-Fi Authentication Flow



## Step-by-Step

### 1. User login:

The user connects to the Wi-Fi network and provides the required credentials.

### 2. Access Request:

The access point or wireless controller sends an authentication request to the RADIUS server.

### 3. Credential verification:

The RADIUS server checks the user's credentials against the configured identity database or authentication system.

### 4. Response:

The RADIUS server sends either **Access-Accept** or **Access-Reject**.

### 5. Network access:

If accepted, the user receives network access according to the configured policies.

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## Question 3

**Describe a real-world scenario from any app you use (such as Zomato, Paytm, or IRCTC) where TACACS+ would be more suitable than RADIUS for managing admin access, and explain why.**

### Answer

#### **Example: Zomato Network Administrators**

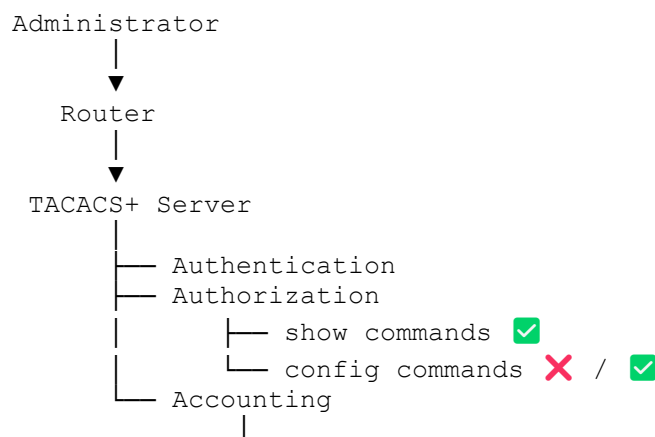
Consider a large company such as Zomato with network administrators managing routers, switches, firewalls, and other infrastructure.

An administrator might need to log in to a router to troubleshoot a network problem.

**TACACS+ would be suitable for this type of administrative access because it supports granular command authorization.** For example, one administrator could be permitted to run monitoring commands such as `show` commands, while a senior network administrator could be authorized to make configuration changes.

TACACS+ can also provide accounting records of administrative activity, helping the organization determine **which administrator executed particular commands and when.**

#### **Why TACACS+?**



**Important:** RADIUS is also capable of authentication and authorization in many network-access scenarios, but TACACS+ is particularly suited to centralized **device-administration and command-level authorization**.

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## Question 4

List two advantages and two disadvantages of using RADIUS for network authentication in a public hotspot (like those found in malls or airports).

Answer

### Advantages of RADIUS

#### 1. Centralized Authentication

RADIUS allows many wireless access points to use a centralized authentication server. This makes it easier to manage user access across a large hotspot.

#### 2. Supports Accounting

RADIUS can record session information such as login time, logout time, and network usage. This can be useful for monitoring and managing public network access.

### Disadvantages of RADIUS

#### 1. Server Dependency

If the RADIUS server or the connectivity to it becomes unavailable, users may not be able to authenticate to the network.

#### 2. Configuration and Maintenance

Deploying and maintaining a secure RADIUS infrastructure requires proper configuration of the RADIUS server, access points, certificates/credentials where applicable, and user database.

### Quick View

| Advantages                                              | Disadvantages                                                                      |
|---------------------------------------------------------|------------------------------------------------------------------------------------|
| RADIUS Centralized authentication<br>Accounting support | Depends on authentication infrastructure<br>Requires configuration and maintenance |

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## Question 5

Use ChatGPT or another AI tool to generate a sample TACACS+ configuration for a Cisco router, then review the output and highlight any commands you would modify for a college lab setup.

**Hint:** Ask the AI tool for “basic TACACS+ configuration for Cisco router” and check for lines like `tacacs-server host` and `aaa new-model`.

### Answer

Below is a **sample Cisco IOS TACACS+ configuration** suitable as a starting point for a college Packet Tracer lab.



## Sample TACACS+ Configuration

### Step 1 — Enable AAA

```
R1> enable
R1# configure terminal
R1(config)# aaa new-model
```

### Step 2 — Configure the TACACS+ Server

For newer IOS syntax:

```
R1(config)# tacacs server TACACS-SERVER
R1(config-server-tacacs)# address ipv4 192.168.100.10
R1(config-server-tacacs)# key cisco123
R1(config-server-tacacs)# exit
```

### Step 3 — Configure AAA Authentication

```
R1(config)# aaa authentication login default group tacacs+ local
```

This means:

**First** → TACACS+ server

**If TACACS+ is unavailable** → local username database

### Step 4 — Create a Local Backup User

```
R1(config)# username admin privilege 15 secret LocalBackup123
```

The local account provides a fallback method if the TACACS+ server cannot be reached.

### Step 5 — Apply Authentication to Console

```
R1(config)# line console 0
R1(config-line)# login authentication default
R1(config-line)# exit
```

## Step 6 — Apply Authentication to VTY Lines

```
R1(config)# line vty 0 4
R1(config-line)# login authentication default
R1(config-line)# transport input ssh
R1(config-line)# exit
```

## Step 7 — Configure TACACS+ Authorization

```
R1(config)# aaa authorization exec default group tacacs+ local
```

## Step 8 — Configure Accounting

```
R1(config)# aaa accounting exec default start-stop group tacacs+
```

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## Commands I Would Modify for a College Lab

| AI-Generated Setting | College Lab Modification                                      | Reason                                                                               |
|----------------------|---------------------------------------------------------------|--------------------------------------------------------------------------------------|
| 192.168.100.10       | <b>Replace with your actual TACACS+ server IP</b>             | The server IP must match the Packet Tracer topology                                  |
| cisco123             | <b>Use the shared secret configured on the TACACS+ server</b> | Both router and server must use the same key                                         |
| LocalBackup123       | <b>Use your own lab-only local password</b>                   | Provides a local fallback account                                                    |
| tacacs+ server group | <b>Check Packet Tracer support</b>                            | Some Packet Tracer versions support only a subset of real Cisco AAA/TACACS+ features |
| line vty 0 4         | <b>Use the VTY range available on your router</b>             | Different router models may provide different VTY lines                              |

### **About tacacs-server host**

Older Cisco IOS configurations commonly use:

```
R1(config)# tacacs-server host 192.168.100.10
R1(config)# tacacs-server key cisco123
```

Newer IOS versions commonly use the named-server syntax:

```
R1(config)# tacacs server TACACS-SERVER
R1(config-server-tacacs)# address ipv4 192.168.100.10
R1(config-server-tacacs)# key cisco123
```

For a Packet Tracer lab, use the syntax supported by your specific router/Packet Tracer version.

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## Final Revision

|  Feature | RADIUS                                          | TACACS+                                             |
|-------------------------------------------------------------------------------------------|-------------------------------------------------|-----------------------------------------------------|
| Transport                                                                                 | UDP                                             | TCP                                                 |
| Authentication Port                                                                       | UDP 1812                                        | TCP 49                                              |
| Accounting Port                                                                           | UDP 1813                                        | TCP 49                                              |
| Encryption                                                                                | Password field protected; not the entire packet | Entire TACACS+ packet body encrypted                |
| Command Authorization                                                                     | Limited/device-dependent                        | Strong granular support                             |
| Common Use                                                                                | Wi-Fi, VPN, network access                      | Network-device administration                       |
| AAA Separation                                                                            | Less granular                                   | Authentication, authorization, accounting separated |



### Key Takeaway

**RADIUS is commonly used for network access such as Wi-Fi and VPN authentication, while TACACS+ is particularly useful for centralized administration of network devices because it provides detailed authorization and accounting capabilities.**